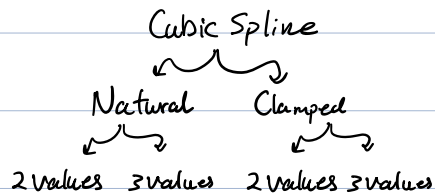


## Cubic Spline Interpolation

→ Types of Cubic Spline &



### ① Construct the natural Cubic Spline &

"b" will given to can find A

⇒ if you have  $x_0, x_1$  only → you'll find  $S(x_0)$

⇒ if you "  $x_0, x_1, x_2$  → you'll find  $S(x_0)$  and  $S(x_1)$

$$S(x_i) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$$

→  $a_i$  and  $x_i$  from Q

$$\rightarrow A^{-1} \cdot b \begin{bmatrix} * \\ * \end{bmatrix} \begin{matrix} \rightarrow c_0 \\ \rightarrow c_1 \end{matrix}$$

$$\rightarrow b_0 = \frac{1}{h_0} (a_1 - a_0) - \frac{h_0}{3} (c_1 + 2c_2)$$

$$\rightarrow d_0 = \frac{1}{3h_0} (c_1 - c_0)$$

$$\rightarrow h_0 = x_1 - x_0$$

### ② Clamped Spline &

- (a)  $S(x)$  is a cubic polynomial, denoted  $S_j(x)$ , on the subinterval  $[x_j, x_{j+1}]$  for each  $j = 0, 1, \dots, n-1$ ;
- (b)  $S_j(x_j) = f(x_j)$  and  $S_j(x_{j+1}) = f(x_{j+1})$  for each  $j = 0, 1, \dots, n-1$ ;
- (c)  $S_{j+1}(x_{j+1}) = S_j(x_{j+1})$  for each  $j = 0, 1, \dots, n-2$ ; (Implied by (b).)
- (d)  $S'_{j+1}(x_{j+1}) = S'_j(x_{j+1})$  for each  $j = 0, 1, \dots, n-2$ ;
- (e)  $S''_{j+1}(x_{j+1}) = S''_j(x_{j+1})$  for each  $j = 0, 1, \dots, n-2$ ;
- (f) One of the following sets of boundary conditions is satisfied:
  - (i)  $S''(x_0) = S''(x_n) = 0$  (natural (or free) boundary);
  - (ii)  $S'(x_0) = f'(x_0)$  and  $S'(x_n) = f'(x_n)$  (clamped boundary).

## ① Natural Cubic Spline 8

Q3: 3. Construct the **natural cubic** spline for the following data.

| a. | $x$         | $f(x)$           |
|----|-------------|------------------|
|    | $x_0 = 8.3$ | $17.56492 = a_0$ |
|    | $x_1 = 8.6$ | $18.50515 = a_1$ |

→ 2 value natural cubic

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{"Identity matrix"}$$

$$A^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

given  $\rightarrow b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$A^{-1} \cdot b = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow \begin{matrix} c_0 \\ c_1 \end{matrix}$$

$$h_0 = 8.6 - 8.3 = 0.3$$

$$\begin{aligned} b_0 &= \frac{1}{h_0} (a_1 - a_0) - \frac{h_0}{3} (c_1 + 2c_0) \\ &= \frac{1}{0.3} (18.50515 - 17.56492) - \frac{0.3}{3} (0 + 2(0)) = 3.1341 \end{aligned}$$

$$\begin{aligned} d_0 &= \frac{1}{3h_0} (c_1 - c_0) \\ &= \frac{1}{3(0.3)} (0 - 0) = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow S(x_0) &= a_0 + b_0(x - x_0) + c_0(x - x_0)^2 + d_0(x - x_0)^3 \\ &= 17.56492 + (3.1341)(x - 8.3) + 0(x - 8.3)^2 + 0(x - 8.3)^3 \\ &= 17.56492 + (3.1341)(x - 8.3) \end{aligned}$$

| b.      | $x$ | $f(x)$             |
|---------|-----|--------------------|
| $x_0 =$ | 0.8 | $0.22363362 = a_0$ |
| $x_1 =$ | 1.0 | $0.65809197 = a_1$ |

$$h_0 = 0.2$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$A^T \cdot b = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \xrightarrow{\text{blue}} \begin{matrix} c_0 \\ c_1 \end{matrix}$$

$$\begin{aligned} b_0 &= \frac{1}{h_0} (a_1 - a_0) - \frac{h_0}{3} (c_1 - 2c_0) \\ &= \frac{1}{0.2} (0.65809197 - 0.22363362) - \frac{0.2}{3} (0 - 2(0)) = 2.17229175 \end{aligned}$$

$$d_0 = \frac{1}{3h_0} (c_1 - c_0) = \frac{1}{3(0.2)} (0 - 0) = 0$$

$$\begin{aligned} S(x_0) &= a_0 + b_0(x - x_0) + d_0(x - x_0)^2 + c_0(x - x_0)^3 \\ &= 0.22363362 + 2.17229175(x - 0.8) + 0(\cancel{x - 0.8})^2 + 0(\cancel{x - 0.8})^3 \\ &= 0.22363362 + 2.17229175(x - 0.8) \end{aligned}$$

| c.            | x | f(x)                         |
|---------------|---|------------------------------|
| $x_0 = -0.5$  |   | $-0.0247500 \rightarrow a_0$ |
| $x_1 = -0.25$ |   | $0.3349375 \rightarrow a_1$  |
| $x_2 = 0$     |   | $1.1010000 \rightarrow a_2$  |

$$h_0 = -0.25 - (-0.5) = 0.25, \quad h_1 = 0 - (-0.25) = 0.25$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ h_0 & 2(h_0+h_1) & h_1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0.25 & 1 & 0.25 \\ 0 & 0 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} 0 \\ \frac{3}{h_1}(a_2-a_1) - \frac{3}{h_0}(a_1-a_0) \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -4.8765 \\ 0 \end{bmatrix}$$

$$A^{-1} \cdot b = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} \Rightarrow A^{-1} = \begin{array}{c} A \\ I \end{array} \begin{bmatrix} 1 & 0 & 0 & | & 1 & 0 & 0 \\ 0.25 & 1 & 0.25 & | & 0 & 1 & 0 \\ 0 & 0 & 1 & | & 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & 0.25 & | & 0.25 & 1 & 0 \\ 0 & 0 & 1 & | & 0 & 0 & 1 \end{bmatrix}$$

$$b_0 = \frac{1}{h_0}(a_1-a_0) - \frac{h_0}{3}(c_1-2c_0) \quad \begin{array}{c} -0.25R_3 + R_1 \rightarrow R_2 \\ I \end{array} \begin{bmatrix} 1 & 0 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & 0 & | & 0.25 & 1 & -0.25 \\ 0 & 0 & 1 & | & 0 & 0 & 1 \end{bmatrix} \quad \begin{array}{c} I \\ A^{-1} \end{array}$$

$$= 1.845125$$

$$b_1 = \frac{1}{h_1}(a_2-a_1) - \frac{h_1}{3}(c_2-2c_1)$$

$$= 2.2515$$

$$A^{-1} \cdot B = \begin{bmatrix} 1 & 0 & 0 \\ 0.25 & 1 & -0.25 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -4.8765 \\ 0 \end{bmatrix}$$

$$d_0 = \frac{1}{3h_0}(c_1-c_0) = -6.502$$

$$= \begin{bmatrix} 0 \\ -4.8765 \\ 0 \end{bmatrix} \begin{array}{l} \rightarrow c_0 \\ \rightarrow c_1 \\ \rightarrow c_2 \end{array}$$

$$d_1 = \frac{1}{3h_1}(c_2-c_1) = 6.502$$

$$s(x_0) = a_0 + b_0(x-x_0) + c_0(x-x_0)^2 + d_0(x-x_0)^3 = -0.0247500 + 1.845125(x+0.5) + 0(x+0.5)^2 + (-6.502)(x+0.5)^3$$

$$s(x_1) = a_1 + b_1(x-x_1) + c_1(x-x_1)^2 + d_1(x-x_1)^3 = 0.3349375 + 2.2515(x+0.25) + (-4.8765)(x+0.25)^2 + 6.502(x+0.25)^3$$

## ② Clamped Cubic Spline

must use these rules

- (a)  $S(x)$  is a cubic polynomial, denoted  $S_j(x)$ , on the subinterval  $[x_j, x_{j+1}]$  for each  $j = 0, 1, \dots, n-1$ ;
- (b)  $S_j(x_j) = f(x_j)$  and  $S_j(x_{j+1}) = f(x_{j+1})$  for each  $j = 0, 1, \dots, n-1$ ;
- (c)  $S_{j+1}(x_{j+1}) = S_j(x_{j+1})$  for each  $j = 0, 1, \dots, n-2$ ; (Implied by (b).)
- (d)  $S'_{j+1}(x_{j+1}) = S'_j(x_{j+1})$  for each  $j = 0, 1, \dots, n-2$ ;
- (e)  $S''_{j+1}(x_{j+1}) = S''_j(x_{j+1})$  for each  $j = 0, 1, \dots, n-2$ ;
- (f) One of the following sets of boundary conditions is satisfied:
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